SUBHAN SAADAT KHAN

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Objective

Master's student in Data Science (AI & ML) at FAU Erlangen with hands-on experience in Python and SQL, exploratory data analysis, and end-to-end ML prototyping. Backend engineering background that includes pulling, cleaning, and joining data from ERP and production databases, building dashboards for non-technical reviewers, and working with Generative AI (GPT-4o, OpenAI API) in real research tools. Seeking a Working Student role in Data Science to support data preparation, EDA, ML prototyping, validation, and visualization for cross-functional teams in a real production setting.

Education

Friedrich-Alexander-Universität (FAU) — **Oct 2024 – Present**
Master's in Data Science (AI & ML) — *Erlangen, Germany*

Monash University — **Feb 2021 – Oct 2023**
Bachelor of Computer Science (Data Science focus) — *Subang Jaya, Malaysia*

Courseworks:

- Data Science
- Data Structures
- Data Analysis
- Object-Oriented Design
- Databases
- Programming Paradigms
- Software Architecture
- Mobile App Development

Technical Skills

Programming Languages: Python (advanced), SQL (advanced), JavaScript, TypeScript, Java, C++, C, HTML/CSS

Data Science & ML: Exploratory Data Analysis (EDA), Feature Engineering, Supervised & Unsupervised Learning, scikit-learn, PyTorch, PyTorch Geometric, TensorFlow, Statistical Methods, Time Series Analysis, Anomaly Detection

Generative AI & LLMs: GPT-4o, OpenAI API, Prompt Engineering, LLM Integration, Human-in-the-Loop AI Workflows

Data Engineering & Tools: Pandas, NumPy, Data Cleaning, ETL/ELT, REST APIs, JSON, ERP / CRM / Production Database Integration, Git, GitHub, Docker, Jupyter

Visualization & Reporting: Power BI, Tableau, Matplotlib, Folium, Interactive Dashboards, Reports for Non-Technical Reviewers

Databases: PostgreSQL, SQL, SQLite, MongoDB, Firebase

Experience

Friedrich-Alexander-Universität — **Jun 2025 – Current**
Working Student / Research Assistant (HiWi) — *Nuremberg, Germany*

- Built a Python data pipeline that pulled, cleaned, validated, and joined 28,000+ records, then exposed the cleaned data through an interactive Folium dashboard for non-technical researchers and stakeholders to review.
- Performed exploratory data analysis and prototyped ML models (Graph Attention Network in PyTorch Geometric) for the HealthPred clinical prediction framework, including feature engineering and evaluation.
- Designed and shipped a GPT-4o-based LLM pipeline integrated into the GAM-Charger model interpretation tool, demonstrating practical Generative AI usage in production-style research tooling.
- Wrote modular, documented code with Git version control and structured experiment tracking, supporting reproducibility and team handover.

ParaLogic — **Jan 2024 – Sept 2024**
Python Developer (ERP & Data Integration) — *Karachi, Pakistan*

- Pulled, cleaned, and joined data from ERP and production databases using Python and SQL, then automated synchronization through third-party REST APIs (JSON), reducing manual data entry by 25%.
- Optimized SQL queries and reporting logic, improving response times for production and financial data by 40% and supporting faster decision-making for chemists, engineers, and finance teams style stakeholders.
- Built and validated automated cost calculations and inventory reports against business logic, reducing accounting discrepancies by 30%.
- Created concise reports and dashboards for non-technical departments, supporting cross-functional review and feedback loops.

Tong Hin **Nov 2022 – Feb 2023**
Software Development Intern *Subang Jaya, Malaysia*

- Designed RESTful APIs in ASP.NET Core for retail order workflows, supporting JSON-based data exchange between UI, backend, and database.
- Modeled relational data structures (ERDs) from business requirements, ensuring scalable and well-documented data architecture.
- Developed a dynamic data-filtering interface, reducing data-entry errors by 20% and improving day-to-day operations.

University Projects

Loan Default Prediction | *Python, scikit-learn, Pandas (EDA & ML Prototyping)*

- Performed exploratory data analysis on credit and loan datasets to identify patterns between credit scores, interest rates, and repayment outcomes.
- Built and evaluated supervised ML models (Decision Tree, Random Forest), achieving 85% accuracy with cross-validation and confusion matrix analysis.
- Translated technical findings into business-readable insights and visualizations for non-technical reviewers.

Efficient Data Stream Anomaly Detection | *Python, NumPy (Unsupervised, Real-Time)*

- Designed an unsupervised anomaly detection algorithm for continuous numerical data streams, optimized for low memory and constant-time updates per data point.
- Handled concept drift and seasonal patterns, applicable to monitoring production process data and KPIs in industrial settings.
- Documented methodology and ensured clean, modular Python code suitable for downstream operationalization.

Time Series Analysis of Mercury Levels | *Python, Pandas, scikit-learn*

- Conducted time series analysis on environmental sensor data, applying decomposition, trend detection, and forecasting techniques.
- Built an end-to-end data preprocessing pipeline including outlier handling, missing-value imputation, and feature engineering.
- Presented findings in clear visualizations and a written summary suitable for non-technical readers.

Extracurricular Activities

TeaMWork Internship Program **Jun 2023 – Jul 2023**
Monash Warwick Alliance *Virtual Internship*

- Conducted structured market research and analysis for an Indonesian cybersecurity firm, identifying key trends and opportunities.
- Co-authored a final report and infographic released to the Indonesian public, communicating technical content to a non-specialist audience.
- Collaborated with international teammates across time zones to deliver structured outputs on a fixed timeline.

Languages

English: Advanced (C1)
German: Elementary (A2, actively improving)